

DSA210 Final Report

The Effect of Time Delay and Platform Triggers on Impulsive Purchasing Behavior

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1. Motivation

With the rapid growth of e-commerce platforms, online purchasing behavior has become increasingly important for both businesses and researchers. Digital shopping environments expose consumers to personalized recommendations, discounts, urgency messages, and fast purchasing systems that may encourage impulsive purchasing behavior. Understanding these behavioral patterns can help businesses improve customer experience while also contributing to the study of consumer decision-making processes.

The motivation behind this project is to explore whether transaction-level purchasing patterns can provide insights into impulsive purchasing behavior. In particular, the project investigates whether time-related variables such as purchase hour, purchase day, and purchasing quantity may help identify impulsive purchases.

I was particularly interested in understanding whether behavioral patterns hidden within transaction data could provide insight into impulsive purchasing tendencies. The project also aims to examine how machine learning techniques can be used to analyze purchasing behavior patterns in online retail environments.

2. Data Source

The dataset used in this project is the Online Retail dataset obtained from Kaggle.

Dataset source:

<https://www.kaggle.com/datasets/vijayuv/onlineretail>

The dataset contains transactional data from a UK-based online retail company. It includes several hundred thousand transactions and contains information such as:

- Invoice number
- Product description
- Quantity
- Invoice date and time
- Unit price
- Customer ID

- Country

The dataset does not directly contain labels for impulsive purchasing behavior. Therefore, an impulsive purchase proxy variable was created using rule-based assumptions related to transaction timing and purchasing behavior patterns.

3. Data Preparation and Feature Engineering

Before analysis, several preprocessing and cleaning steps were applied to the dataset.

The following operations were performed:

- Removed missing values
- Removed invalid or negative quantities
- Converted the invoice date column into datetime format
- Extracted hour and day information from transaction timestamps
- Created weekday and weekend indicators
- Created time-based purchasing features

Feature engineering played an important role in preparing the dataset for exploratory analysis and machine learning models. Additional features such as purchase hour and purchase day were used to analyze temporal purchasing behavior patterns.

Since the dataset did not contain a direct impulsive purchase label, a proxy label was created based on assumptions related to transaction timing and purchasing conditions. Although this proxy does not directly measure impulsiveness, it provides an approximate representation for analysis purposes.

4. Exploratory Data Analysis (EDA)

Several exploratory data analysis techniques and visualizations were applied to better understand purchasing behavior patterns.

The analysis included:

- Purchases by hour
- Purchases by day
- Quantity distribution
- Top purchasing countries
- Average quantity by hour

The visualizations showed that purchasing activity is concentrated during certain hours of the day. Transaction frequency also varies across different days of the week. The majority of transactions involve relatively small quantities, while a smaller number of transactions contain very large quantities.

Country-based analysis showed that transaction activity is geographically concentrated, particularly around the United Kingdom and several European countries.

The EDA results suggest that temporal purchasing behavior may contain useful signals for identifying impulsive purchasing tendencies.

5. Hypothesis Testing

Several hypothesis tests were conducted to examine whether purchasing behavior changes across different time periods.

Hypothesis Test 1: Daytime vs Evening Purchases

- **Null Hypothesis (H0):** There is no significant difference between daytime and evening purchase quantities.
- **Alternative Hypothesis (H1):** There is a significant difference between daytime and evening purchase quantities.

A t-test was applied to compare purchasing quantities across different time periods.

Hypothesis Test 2: Weekday vs Weekend Purchases

- **Null Hypothesis (H0):** There is no significant difference between weekday and weekend purchasing behavior.
- **Alternative Hypothesis (H1):** There is a significant difference between weekday and weekend purchasing behavior.

The p-value obtained from the analysis indicated a statistically significant difference between weekday and weekend purchasing quantities.

Hypothesis Test 3: Morning vs Evening Purchases

- **Null Hypothesis (H0):** There is no significant difference between morning and evening purchase quantities.
- **Alternative Hypothesis (H1):** There is a significant difference between morning and evening purchase quantities.

The results did not provide strong statistical evidence of a difference between morning and evening purchasing behavior.

These hypothesis tests helped support the exploratory findings obtained from the visualizations.

6. Machine Learning Methods

Several machine learning models were applied to classify purchases as impulsive or non-impulsive based on engineered transaction-level features.

The models used in this project include:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

The dataset was divided into training and test datasets using train-test split methodology.

To improve model performance, hyperparameter tuning was performed using GridSearchCV. Different parameter combinations were tested for each model.

The models were evaluated using multiple performance metrics:

- Accuracy
- Precision
- Recall
- F1 Score

In addition, visualizations were created to compare:

- Model performance metrics
- Training and test performance
- Validation performance from GridSearchCV

These comparisons helped evaluate model generalization and potential overfitting behavior.

7. Results and Findings

The machine learning results showed that time-related and transaction-level features may help identify purchasing behavior patterns associated with impulsive purchases.

Decision Tree and Random Forest models achieved very high performance scores, while Logistic Regression produced relatively lower performance. However, these results should be interpreted carefully because the dataset is highly imbalanced and the impulsive purchase label is based on a rule-based proxy rather than direct behavioral observations.

The comparison between training, validation, and test scores provided additional insight into model performance and generalization behavior.

The project findings suggest that transaction timing, quantity, and purchasing conditions may contribute to identifying impulsive purchasing tendencies in online retail environments.

8. Limitations and Future Work

This project has several limitations.

First, the dataset does not directly measure impulsive purchasing behavior. Instead, impulsive purchases were approximated using a rule-based proxy variable. Therefore, the analysis may not fully capture the psychological aspects of impulsive buying behavior.

Second, the dataset mainly contains transactional information and does not include additional behavioral variables such as:

- Customer browsing history
- Session duration
- Clickstream behavior
- Exposure to discounts or advertisements
- Emotional or psychological factors

Future work may include:

- Collecting richer behavioral datasets
- Using customer interaction data
- Applying advanced machine learning techniques
- Exploring deep learning methods
- Incorporating recommendation system data and platform trigger variables

These improvements could provide a more realistic understanding of impulsive purchasing behavior.

9. Conclusion

This project explored impulsive purchasing behavior using the Online Retail dataset. Through exploratory data analysis, hypothesis testing, and machine learning methods, several patterns related to purchasing time and quantity were identified.

Although the dataset does not directly measure impulsiveness, the analysis demonstrates how transaction-level features can be used to study customer purchasing behavior. The results suggest that temporal and transactional patterns may contribute to understanding impulsive purchasing tendencies in e-commerce environments.

Overall, the project demonstrates how data science techniques can be applied to analyze online consumer behavior and investigate behavioral patterns using real-world retail transaction data.

10. AI Assistance Disclosure

AI-based tools were used during the development of this project for guidance related to coding, debugging, writing assistance, and interpretation of analysis results. All final

decisions, implementations, and project organization were reviewed and completed by the author.